

SMYRNA, TN, USA

WMO: 723273

Lat: 36.009N

Lon: 86.520W

Elev: 166

StdP: 99.35

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 13827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.6	-7.2	-15.6	1.0	-6.4	-13.6	1.2	-5.2	10.1	5.0	9.0	7.5	2.3	0	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	37.2	24.6	35.5	24.4	34.1	24.1	26.3	33.5	25.7	32.5	25.2	31.7	3.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	19.9	29.1	23.9	19.2	28.5	23.4	18.6	27.9	82.4	33.3	79.9	32.4	77.5	31.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.6	6.8	-12.8	39.0	2.8	3.1	-14.8	41.2	-16.4	43.0	-18.0	44.8	-20.1	47.0		
			-13.5	27.5	2.6	0.7	-15.4	28.0	-16.9	28.4	-18.4	28.7	-20.3	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	3.6	6.0	11.0	15.8	20.9	25.6	27.0	26.7	23.2	16.3	9.2	6.2	(6)
(7)		DBStd	9.49	6.23	6.19	5.73	4.75	4.03	2.64	2.39	2.64	3.55	4.91	4.90	5.50	(7)
(8)		HDD10.0	647	215	142	58	9	0	0	0	0	7	71	146		(8)
(9)		HDD18.3	1920	457	347	233	103	24	0	0	6	99	275	376		(9)
(10)		CDD10.0	2841	17	29	90	182	339	467	527	517	395	202	48	28	(10)
(11)		CDD18.3	1073	0	1	7	26	105	217	269	259	151	36	2	1	(11)
(12)		CDH23.3	11392	0	5	94	339	1071	2316	2917	2759	1455	420	14	1	(12)
(13)		CDH26.7	5401	0	0	19	82	416	1153	1491	1444	661	134	0	0	(13)
(14)	Wind	WSAvg	2.5	3.0	3.2	3.2	3.0	2.4	2.2	2.0	2.0	2.3	2.4	2.8	(14)	
(15)	Precipitation	PrecAvg	1294	127	109	126	113	126	100	112	90	89	82	91	129	(15)
(16)		PrecMax	1545	249	253	268	201	397	271	212	217	221	209	157	313	(16)
(17)		PrecMin	978	54	52	70	22	13	2	52	35	5	0	15	60	(17)
(18)		PrecStd	145	41	47	47	51	72	55	44	45	52	48	39	58	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	24.1	29.0	31.1	34.5	37.9	39.1	40.2	36.7	33.0	25.3	22.9	(19)
(20)		MCWB	15.9	17.1	18.7	19.7	23.5	23.2	24.9	24.5	23.5	20.3	16.2	17.8	(20)	
(21)		2%	DB	18.7	21.4	26.2	29.1	32.3	35.8	36.5	37.6	34.3	30.1	22.7	20.2	(21)
(22)		MCWB	15.1	16.1	17.2	18.8	21.9	24.0	25.1	24.5	22.9	20.3	15.9	16.4	(22)	
(23)		5%	DB	16.6	18.9	23.4	27.1	30.8	34.2	34.8	35.5	32.7	27.9	20.5	17.9	(23)
(24)		MCWB	13.7	14.4	15.7	18.1	21.4	23.9	25.0	24.5	22.8	19.5	15.1	14.8	(24)	
(25)		10%	DB	14.1	16.6	20.8	25.2	29.1	32.7	33.5	33.5	30.8	25.6	18.3	15.5	(25)
(26)		MCWB	10.8	12.1	14.5	17.2	20.9	23.2	24.6	23.7	21.9	18.3	14.0	12.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	18.7	19.7	21.9	24.4	26.3	27.2	27.1	25.4	23.9	19.6	19.2	(27)
(28)		MCDWB	19.8	22.2	26.3	28.6	31.9	33.9	34.8	35.4	31.6	29.3	22.4	21.4	(28)	
(29)		2%	WB	15.7	17.5	18.3	20.6	23.4	25.7	26.5	26.1	24.7	22.6	18.0	17.3	(29)
(30)		MCDWB	17.9	20.4	24.4	26.3	30.2	33.0	33.5	33.4	30.8	27.1	20.5	19.5	(30)	
(31)		5%	WB	14.0	15.5	17.1	19.4	22.7	25.0	25.9	25.4	24.0	21.1	16.4	15.4	(31)
(32)		MCDWB	16.4	18.4	21.8	24.8	28.8	31.9	32.6	31.9	29.8	25.2	18.8	17.3	(32)	
(33)		10%	WB	11.6	13.3	15.7	18.2	21.8	24.3	25.3	24.8	23.2	19.7	14.7	13.0	(33)
(34)		MCDWB	13.5	15.4	19.6	23.2	27.4	30.3	31.9	30.9	28.3	24.0	17.9	15.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.8	11.7	12.9	14.0	12.8	13.0	12.2	12.8	13.1	14.1	13.3	10.8	(35)	
(36)		5% DB	MCDBR	12.7	14.5	16.5	16.3	15.3	15.2	14.2	15.7	15.6	16.6	16.3	12.9	(36)
(37)		MCWBR	9.6	9.7	9.0	7.5	6.1	5.0	4.4	4.7	5.4	7.6	9.9	9.3	(37)	
(38)		5% WB	MCDBR	11.3	11.7	13.7	13.0	12.7	12.7	12.0	12.3	12.2	13.0	11.1	11.1	(38)
(39)		MCWBR	9.6	9.0	8.3	6.9	5.7	5.0	4.3	4.5	4.8	6.9	8.4	9.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.328	0.373	0.402	0.454	0.480	0.513	0.527	0.431	0.377	0.337	0.312	(40)	
(41)		taud	2.477	2.447	2.354	2.269	2.169	2.132	2.060	2.006	2.262	2.398	2.463	2.515	(41)	
(42)		Ebn at Noon	877	912	894	881	837	812	782	761	823	843	848	864	(42)	
(43)		Edh at Noon	84	97	116	133	150	155	166	171	126	101	85	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.22	2.79	3.91	5.12	5.71	6.23	5.93	5.48	4.67	3.64	2.62	1.91	(44)	
(45)		RadStd	0.19	0.35	0.37	0.36	0.47	0.42	0.31	0.27	0.40	0.46	0.28	0.18	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (22 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+98	(47)

Nomenclature: See separate page